

1. (30 points) Check if the vectors $\{v_1 = (1, -1, 3, -1), v_2 = (1, -1, 4, 2), v_3 = (1, -1, 5, 7)\}$ of $\mathbb{R}^4$ linearly dependent or independent.

$$\text{Let } \alpha_1 (1, -1, 3, -1) + \alpha_2 (1, -1, 4, 2) + \alpha_3 (1, -1, 5, 7) = (0, 0, 0, 0)$$

Then $\alpha_1 + \alpha_2 + \alpha_3 = 0$
 $-\alpha_1 - \alpha_2 - \alpha_3 = 0$ ← clearly this is redundant. You may continue without this or well.
 $3\alpha_1 + 4\alpha_2 + 5\alpha_3 = 0$
 $-\alpha_1 + 2\alpha_2 + 7\alpha_3 = 0$

Corresponding augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -1 & -1 & -1 & 0 \\ 3 & 4 & 5 & 0 \\ -1 & 2 & 7 & 0 \end{array} \right] \begin{array}{l} R_1 + R_2 \rightarrow R_2 \\ -3R_1 + R_3 \rightarrow R_3 \\ R_1 + R_4 \rightarrow R_4 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 8 & 0 \end{array} \right] \begin{array}{l} R_2 \leftrightarrow R_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 3 & 8 & 0 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_4} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-3R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{\frac{1}{2}R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0 \text{ is the unique soln.}$$

2. (35 points) Let $\mathbb{P}_2$ be the vector space of all polynomials of degree less than or equal to 2 with real coefficients and let W be the subset of $\mathbb{P}_2$ containing the polynomials that are divisible by $x - 2$. Determine whether W is a subspace of $\mathbb{P}_2$. If not, give at least one property that is not satisfied.

(i) Since $p(x) = 0$ is divisible by $(x - 2)$,
 $p(x) \in W$ and $W \neq \emptyset$.

(ii) $\forall p_1(x)$ and $p_2(x)$ in W , since they are
 divisible by $(x - 2)$, $p_1(x) = (x - 2)q_1(x)$
 $p_2(x) = (x - 2)q_2(x)$

Then, $p_1(x) + p_2(x) = (x - 2)q_1(x) + (x - 2)q_2(x)$
 $= (x - 2)(q_1(x) + q_2(x))$ is
 divisible by $(x - 2)$

$\Rightarrow p_1(x) + p_2(x) \in W \Rightarrow W$ is closed under addition

(iii) $\forall p(x) \in W$, $p(x) = (x - 2)q(x)$ and $\forall c \in \mathbb{R}$,
 $cp(x) = (x - 2)\underbrace{cq(x)}_{\text{poly of degree } \leq 1}$ is divisible by $(x - 2)$.

$\Rightarrow cp(x) \in W \Rightarrow W$ is closed under scalar
 mult.

Then W is a subspace of $\mathbb{P}_2$.

3. (35 points) (a) Let V be a finite dimensional vector space and S be a subspace of V . Show that every basis for S is contained in some basis for V .

(b) In $\mathbb{R}^3$, given $\underline{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\underline{u}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, find $\underline{u}_3$ that is orthogonal to $\underline{u}_1$ and $\underline{u}_2$, such that $\{\underline{u}_1, \underline{u}_2, \underline{u}_3\}$ is a basis for $\mathbb{R}^3$

(a) (question from the problem set 3)

If $S \neq V$, and $\{s_1, s_2, \dots, s_k\}$ is a basis of S , $\exists y \in V$ s.t. $y \notin S$ then $y \notin \text{span}\{s_1, s_2, \dots, s_k\} \Rightarrow \{s_1, s_2, \dots, s_k, y\}$ is L.I. If $\text{span}\{s_1, s_2, \dots, s_k, y\} = V$, then this is a basis of V , if not, $\exists y_2 \in V$ s.t. $y_2 \notin \text{span}\{s_1, \dots, s_k, y\} \Rightarrow \{s_1, \dots, s_k, y_1, y_2\}$ is L.I. If it spans V , it is the basis we are looking for. If not, we can continue similarly and extend it to a basis of V (since V is finite dim'ed).

(b) observe that $\{u_1, u_2\}$ is L.I. Then taking a vector $u \notin \text{span}\{u_1, u_2\}$ and finding the projection error, we can obtain a vector $u_3 = u - \text{proj}_{\text{span}\{u_1, u_2\}} u$ we will be done. OR we are looking for $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \notin \text{span}\{u_1, u_2\}$

s.t. $\begin{cases} x + y = 0 \\ x - z = 0 \end{cases}$ so that $\begin{cases} u_3 \cdot u_1 = 0 \\ u_3 \cdot u_2 = 0 \end{cases}$

Then $z = \alpha, x = \alpha, y = -\alpha$. i.e. $\alpha \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ is orthogonal

to both u_1 and u_2 . $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$ is L.I. hence a basis for $\mathbb{R}^3$.